System Selection & Problem Statement

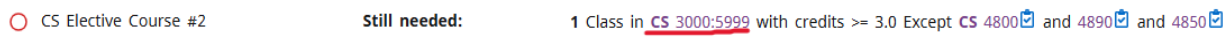
Platform Name: Weber State University CatTracks (Degree Evaluation & Course Planning Module)

Target User Demographics: Undergraduate Computer Science students at Weber State University balancing course requirements, work schedules, and graduation timelines.

Core User Tasks: Navigating degree evaluations to identify elective requirements like selecting 2 courses from CS 3000:5999, reviewing available electives, and choosing classes for upcoming semesters.

Business & Human Consequences: When students click the elective link(CS 3000:5999), a non-responsive modal window opens that dumps an unorganized, lengthy list of every historical and active course matching that code range. Because unavailable courses are mixed in and no side-by-side comparison tool exists, students are forced to perform exhausting vertical scrolling or take multiple manual screenshots just to compare classes. This high cognitive load increases task completion time, creates decision fatigue, and often forces students to seek unnecessary advising appointments for simple course checks.

Figure 1: CatTracks degree evaluation view highlighting the clickable CS 3000:5999 text link for CS Elective requirements.



Qualitative Contextual Research

1. User Research Protocol

- **Objective:** Observe how CS students navigate degree evaluations to select upper-division electives, identifying where cognitive load spikes during comparison.
- **Participant Demographics:** Junior/Senior CS undergraduate students at WSU who have actively registered or plan to register their electives.
- **Test Scenario:** "Open your CatTracks, find your remaining CS elective requirements, and try to pick two upper-division classes that fit your schedule for next semester.".

2. Interview Script

- **Introduction**
    - **Interviewer:** "Thank you for taking a few minutes to help me with this research. Today, I will observe how students figure out their Computer Science upper division electives in CatTracks. There are no right or wrong answers, I'm just observing how you normally navigate the system."
    - **Question 1:** "Can you walk me through the last time you tried to figure out which CS elective courses you needed or were available for an upcoming semester?"
    - **Question 2:** "When looking at your degree requirements, how do you typically figure out what 'CS 3000:5999' actually entails?"
- **Observation**
    - **Interviewer:** "Now, I'd like you to open your CatTracks degree evaluation, scroll down to your CS Elective Course requirement, and click on the purple link that says CS 3000:5999. As you look through the pop-up window, please think out loud and tell me what you're looking at, what you're clicking, and any thoughts or frustrations you have."
- **Probing & Debrief Questions**
    - **Interviewer:** "While you're looking at that list, how do you feel about the way courses are displayed in this pop-up window?"
        - **Probing Follow-up:** "What do you notice about the classes that aren't available?"
    - **Interviewer:** "As you try to weigh your options, how are you keeping track of the differences between the course descriptions or meeting times while scrolling?"
        - **Probing Follow-up:** "Do you feel like you have enough information on your screen at one time to easily make a choice, or do you feel like you have to jump back and forth?"
- **Wrap-up**
    - **Interviewer:** "To wrap things up, if you can change how this elective selection window works in CatTracks, what is the single biggest improvement you would make?"

- **Interviewer:** "Thank you so much for your time and feedback!"

3. Observation Notes & Findings

During the interview, participants exhibited immediate cognitive friction and task disruption upon opening the CS 3000:5999 course information pop-up window. When prompted about how courses are displayed, participants described the popup as a dense, unorganized text dump that mixes active courses with unavailable courses.

Regarding how they tracked course details, participants reported significant frustration, because the interface lacks a side-by-side comparison tool or filter options. Users were forced to continuously scroll up and down between course descriptions, prerequisites, and meeting time tables. One participant noted having to take screenshots to visually compare two classes side-by-side.

Key Takeaways:

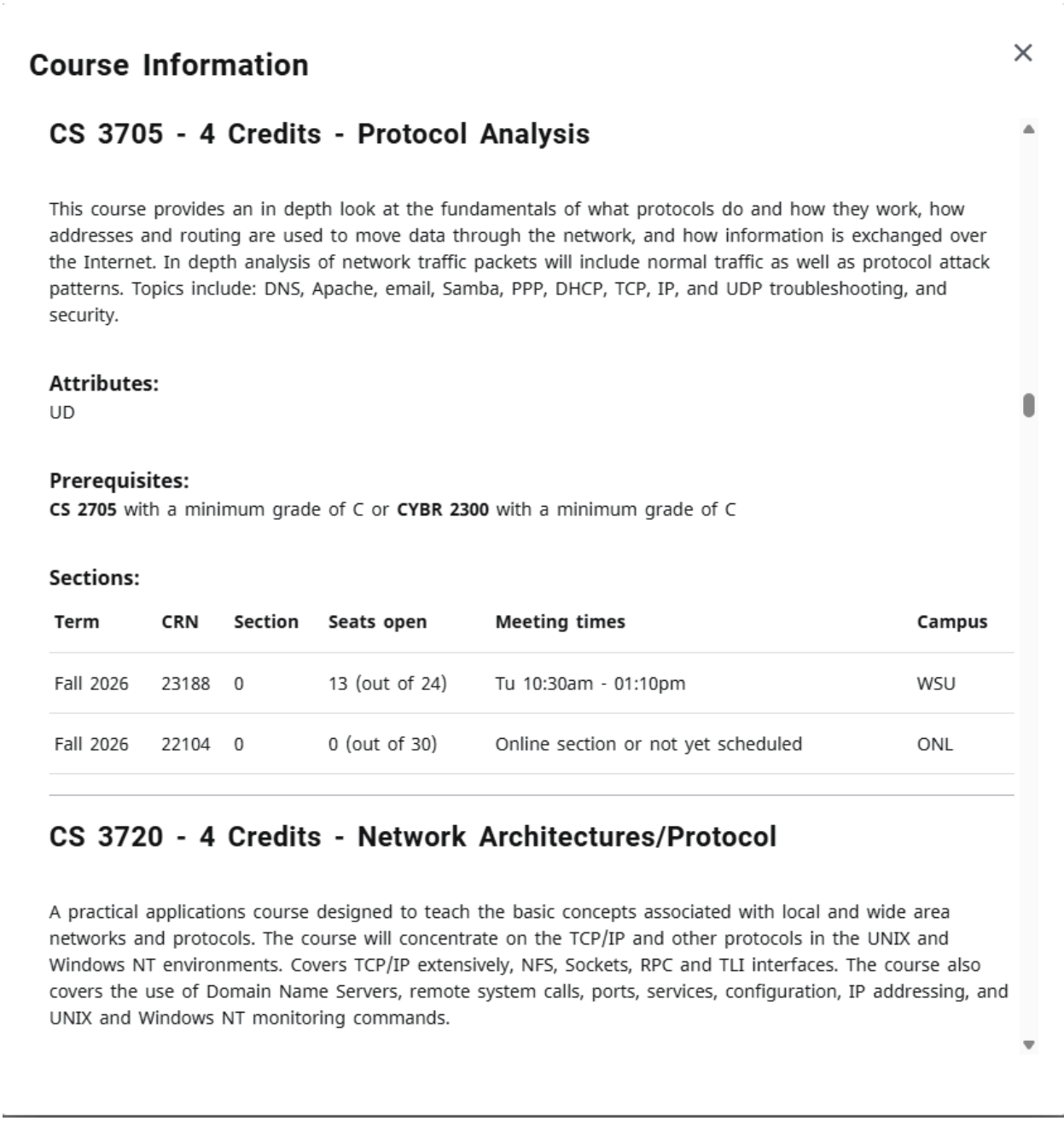
- **Mental Model Mismatch:** Students expect a degree evaluation popup to function like an interactive, filterable course catalog.
- **Task Breakdown Point:** Excessive vertical scrolling and lack of visual hierarchy caused users to lose their place when cross-referencing course prerequisites with available section times.
- **Direct User Quote:** "Why are classes that aren't offered mixed right in with the ones I can actually take? It just creates clutter and makes me scroll a lot."

Heuristic Evaluation & Accessibility Audit

Heuristics Audit(Using Nielsen's 10 Usability Heuristics):

- **Flexibility & Efficiency of Use (Severity 3 - Major Usability Problem):** The interface lacks filters or accelerators. Students cannot filter out courses that aren't offered in the target semester, forcing them to manually scan through irrelevant data.

  Figure 2: Course Information modal popup displaying mixed unavailable/active courses and forcing deep vertical scrolling without filtering or comparison tools.



Course Information

CS 3705 - 4 Credits - Protocol Analysis

This course provides an in depth look at the fundamentals of what protocols do and how they work, how addresses and routing are used to move data through the network, and how information is exchanged over the Internet. In depth analysis of network traffic packets will include normal traffic as well as protocol attack patterns. Topics include: DNS, Apache, email, Samba, PPP, DHCP, TCP, IP, and UDP troubleshooting, and security.

Attributes:
UD

Prerequisites:
CS 2705 with a minimum grade of C or **CYBR 2300** with a minimum grade of C

Sections:

Term	CRN	Section	Seats open	Meeting times	Campus
Fall 2026	23188	0	13 (out of 24)	Tu 10:30am - 01:10pm	WSU
Fall 2026	22104	0	0 (out of 30)	Online section or not yet scheduled	ONL

CS 3720 - 4 Credits - Network Architectures/Protocol

A practical applications course designed to teach the basic concepts associated with local and wide area networks and protocols. The course will concentrate on the TCP/IP and other protocols in the UNIX and Windows NT environments. Covers TCP/IP extensively, NFS, Sockets, RPC and TLI interfaces. The course also covers the use of Domain Name Servers, remote system calls, ports, services, configuration, IP addressing, and UNIX and Windows NT monitoring commands.

- **Recognition Rather Than Recall (Severity 2 - Minor Usability Problem):** Because users cannot compare courses side-by-side, they must hold information in their working memory or take external screenshots to remember details of Class A while scrolling down to look at Class B.

- **Aesthetic & Minimalist Design (Severity 2 - Minor Usability Problem):** The popup window presents a dense block of text and multiple sections without clear visual hierarchy or data tables, cluttering the view with redundant or inactive course data.

Baseline Accessibility (WCAG 2.1 AA) Audit:

To evaluate the Weber State University CatTracks course lookup modal window for inclusive design compliance, both automated inspection tools (axe DevTools / WAVE) and manual browser evaluation protocols were used across three core technical accessibility vectors.

Visual Color Contrast Ratios (WCAG SC 1.4.3):

- **Link Text Contrast:** The interactive purple text used for elective links (CS 3000:5999) measures at a 3.8:1 contrast ratio against the white background, failing the minimum 4.5:1 threshold.

- **Secondary Metadata:** The muted gray labels within the section tables (such as "Seats open" or "Online section or not yet scheduled") fall below 3.2:1, making secondary information difficult to decipher.

Keyboard Navigation & Visible Focus Indicators (WCAG SC 2.4.7):

- **Modal Focus State:** Tabbing onto the CS 3000:5999 link correctly opens the popup modal. However, the close button (X icon in the upper right) strips away standard outline styling without providing an alternative high-contrast outline indicator.

- **Focus Trapping:** While focus remains within the modal window, scrolling the long inner container via keyboard requires multiple tab presses to reach hidden overflow elements, causing navigation fatigue.

Semantic Structure & ARIA Modal Attributes (WCAG SC 1.3.1 & 4.1.2):

- **Missing ARIA Roles:** The course details popup is rendered as a visual CSS layer without proper role="dialog" or aria-modal="true" markup.

- **Unstructured Table Data:** Course section lists (Term, CRN, Section, Seats Open, Meeting Times) lack accessible table headers (<th scope="col">), preventing screen readers from announcing column contexts as users navigate through rows.

Post-Mortem Reflection (Metacognition)

How did combining qualitative user observations with an expert heuristic evaluation reveal usability flaws that a developer testing their own code would likely overlook?

Developers testing their own code often have system familiarity bias. When a developer builds or tests an interface like the CatTracks elective course lookup modal, they already understand the underlying data structure, know exactly where to click, and have a mental model perfectly aligned with the system architecture. Consequently, developer testing focuses primarily on functional verification, checking whether clicking CS 3000:5999 successfully opens a modal populated with database records.

Combining an expert heuristic evaluation with qualitative user observations exposed critical breakdowns that functional testing hides:

- **Uncovering Cognitive Overload:** While a developer sees a functioning scrollable list, observing real students revealed how the lack of filtering created severe cognitive fatigue. Users struggled to retain details for Class A while scrolling past inactive courses to inspect Class B, a friction point that heuristic 6 (Recognition Rather Than Recall) identifies but code tests ignore.

- **Exposing Workarounds and Mental Model Mismatches:** Observing real students revealed "guerrilla workarounds", such as taking screenshots to visually cross-reference course schedules. A developer assumes users read top-to-bottom but observation proved users rapidly scan, lose their place due to poor visual hierarchy, and get frustrated by courses that aren’t available mixed with active courses.

Empirical qualitative data anchors design decisions in actual user behavior rather than developer assumptions, proving that software can be 100% bug-free functionally while remaining unusable humanely.

Identify one heuristic violation that directly creates a legal or ethical accessibility barrier under WCAG 2.1 AA, and explain how resolving that violation improves the user experience for both disabled users and the general public.

The violation of Nielsen's Heuristic #4 (Consistency & Standards) and Heuristic #7 (Flexibility & Efficiency of Use) in the modal popup directly triggers a severe accessibility failure under WCAG 2.1 AA Success Criterion 2.4.7: Focus Visible (Level AA) and SC 1.3.1: Info and Relationships (Level A).

Specifically, the modal popup strips away standard visible focus indicators for keyboard navigation and fails to implement proper semantic role="dialog" attributes.

Legal & Ethical Barriers:

- **Ethical Implication:** Public and educational institutions receiving federal funding (such as Weber State University) are legally obligated under Section 504/508 of the Rehabilitation Act and Title II of the ADA to provide equal access to educational tools.
- **Accessibility Exclusion:** A blind student using a screen reader or a student with a physical tremor navigating via keyboard cannot determine when the modal opens or where keyboard focus is located. This creates an absolute barrier, preventing them from independently selecting required courses to graduate.

Universal Benefit

Resolving this violation by adding high-contrast :focus-visible outline rings, semantic ARIA attributes, and clear visual boundaries, creates a curb-cut effect that benefits all users:

- For Disabled Users: Screen reader users receive clear auditory cues ("Course Information dialog window opened"), and keyboard-only users can seamlessly tab through course section rows with full visual feedback.

- For the General Public: Clear focus indicators and structured modal boundaries benefit power users navigating via keyboard shortcuts, students working on laptops with trackpads disabled, users working in bright sunlight where screen visibility is washed out, and users multi-tasking across windows during high-stress registration windows.